

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 1: Real Numbers

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 HCF and LCM by prime factorisation

1. Let m be the HCF and n the LCM of 98 and 28. Find the value of $n - 7m$. **CBSE 2025, 1 mark**

SOLUTION

$$98 = 2 \times 7^2 \text{ and } 28 = 2^2 \times 7$$

$$\text{HCF } m = 2 \times 7 = 14 \text{ and LCM } n = 2^2 \times 7^2 = 196$$

$$n - 7m = 196 - 98$$

Answer: 98

2. Two positive integers are $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are primes. What is the LCM of p and q ? (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ **CBSE 2024, 1 mark**

SOLUTION

$$p = 2 \times 3^2 \times a^2 \times b^4 \text{ and } q = 2^2 \times 5 \times a^3 \times b^2$$

$$\text{LCM takes the greatest power of every prime: } 2^2 \times 3^2 \times 5 \times a^3 \times b^4 = 180a^3b^4$$

Answer: (d)

3. At a workshop there are 48 French teachers, 80 Hindi teachers and 144 English teachers. Every room must hold the same number of teachers, and all the teachers in a room must teach the same subject. What is the least number of rooms needed? **CBSE 2024, 3 marks**

SOLUTION

For the fewest rooms, each room must seat the greatest possible number, which is the HCF of 48, 80 and 144.

$$48 = 2^4 \times 3, 80 = 2^4 \times 5, 144 = 2^4 \times 3^2, \text{ so HCF} = 2^4 = 16$$

$$\text{Number of rooms} = \frac{48}{16} + \frac{80}{16} + \frac{144}{16} = 3 + 5 + 9$$

Answer: 17 rooms

2 The Fundamental Theorem of Arithmetic

4. For which natural numbers n is $(-1)^n + (-1)^8 = 0$? **CBSE 2025, 1 mark**

SOLUTION

$(-1)^8 = 1$, because the power is even.

So we need $(-1)^n = -1$, which happens only when the power is odd.

Answer: n is any odd natural number

5. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. **CBSE 2024, 2 marks**

SOLUTION

$$5 \times 11 \times 17 + 3 \times 11 = 11(5 \times 17 + 3) = 11 \times 88$$

The number has factors other than 1 and itself (for example 11 and 88).

Hence it is a composite number.

3 Irrational numbers and their proofs

6. Which of these is a rational number lying between $\sqrt{3}$ and $\sqrt{5}$? (a) 1.4142387954012... (b) $2.32\bar{6}$ (c) π (d) 1.857142 **CBSE 2025, 1 mark**

SOLUTION

$\sqrt{3} \approx 1.732$ and $\sqrt{5} \approx 2.236$, so the number must lie between these two values.

(a) is less than 1.732, and it neither ends nor repeats, so it is irrational. (b) is rational but greater than 2.236. (c) $\pi \approx 3.14$ is irrational and too large.

(d) 1.857142 is a terminating decimal, so it is rational, and $1.732 < 1.857142 < 2.236$.

Answer: (d)

7. Show that $\frac{1}{\sqrt{5}}$ is an irrational number. **CBSE 2025, 3 marks**

SOLUTION

Suppose $\frac{1}{\sqrt{5}}$ is rational. Then $\frac{1}{\sqrt{5}} = \frac{a}{b}$, where a and b are positive integers with no common factor other than 1.

Cross-multiplying, $b = \sqrt{5}a$. Squaring both sides, $b^2 = 5a^2$.

So 5 divides b^2 , and therefore 5 divides b . Write $b = 5c$.

Then $25c^2 = 5a^2$, so $a^2 = 5c^2$. So 5 divides a^2 , and therefore 5 divides a .

Now 5 is a common factor of a and b . This contradicts the fact that a and b have no common factor other than 1.

So our assumption is wrong, and $\frac{1}{\sqrt{5}}$ is irrational.

8. Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. **CBSE 2024, 2 marks**

SOLUTION

Assume, to the contrary, that $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging: $\sqrt{3} = \frac{5b - a}{2b}$

Since a and b are integers, the right side is rational, so $\sqrt{3}$ is rational. This contradicts the given fact that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.

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